

CSE 206 (Digital Logic Design Sessional)

Experiment No. 7

Name of the Experiment-

Flip-Flops and Registers

Submitted by-

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CSE

Level 2 Term 1

Problem 1

Statement:

We have to design and implement a master-slave JK flip-flop using only NAND gates.

Excitation Table:

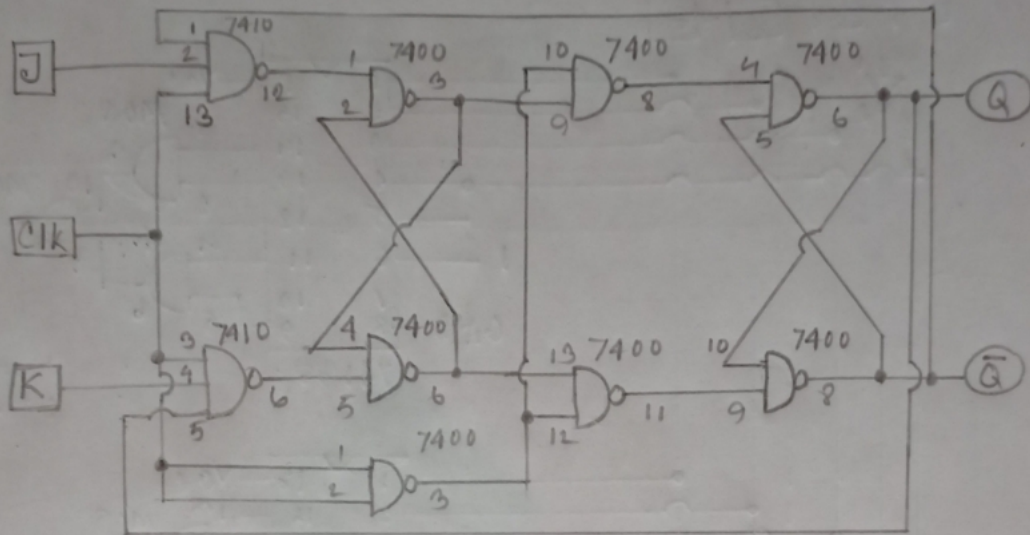
CLK	J	K	Q_n	Q_{n+1}	Mode
↓	X	X	0	0	No change
↓	X	X	1	1	
↑	0	0	0	0	No change
↑	0	0	1	1	
↑	0	1	0	0	Reset
↑	0	1	1	0	
↑	1	0	0	1	Set
↑	1	0	1	1	
↑	1	1	0	1	Toggle
↑	1	1	1	0	

Equation:

J	KQ_n			
	00	01	11	10
0	0	1	0	0
1	1	1	0	1

$$Q_{n+1} = \bar{K}Q_n + J\bar{Q}_n$$

Circuit Diagram:



Problem 2

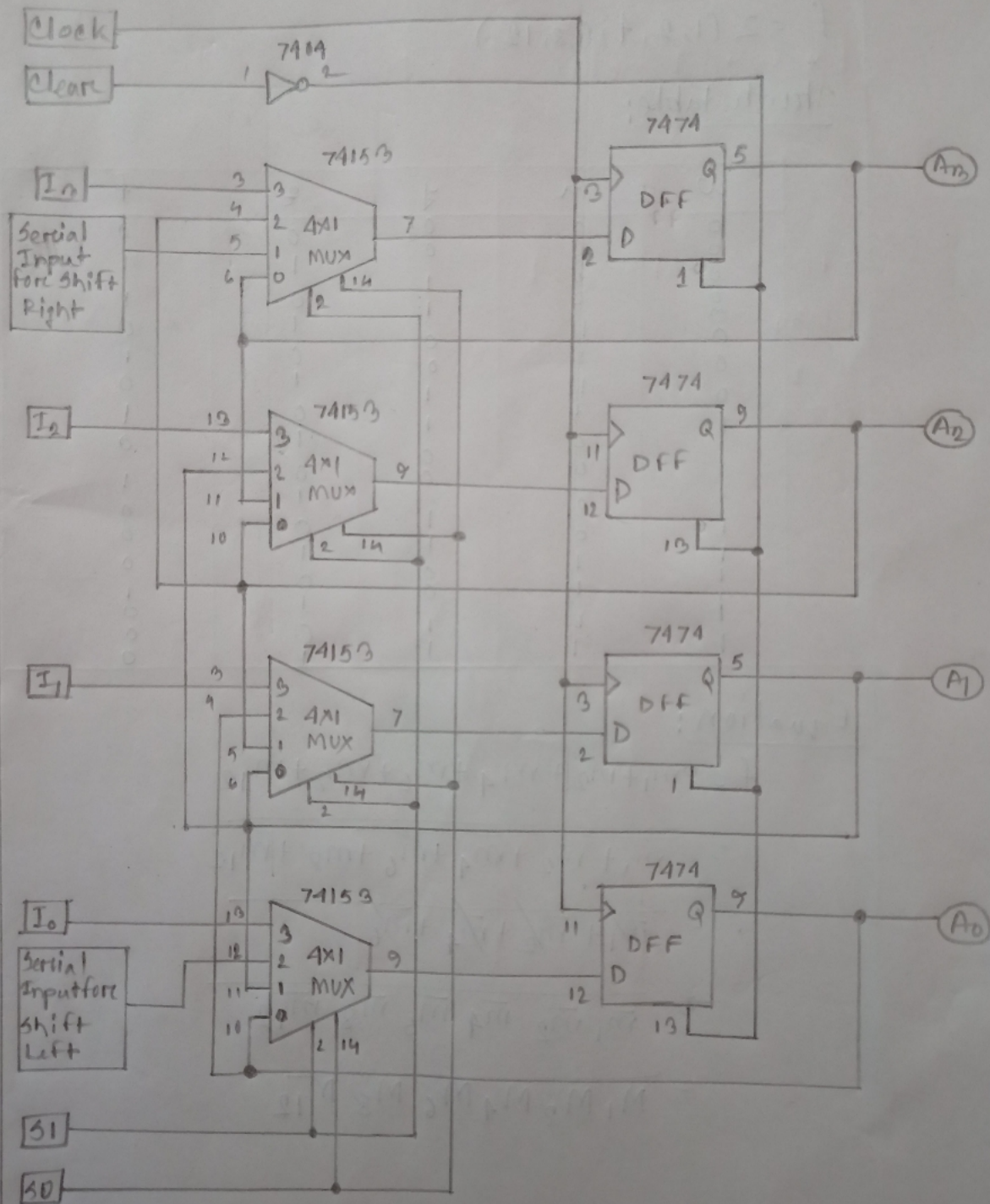
Statement:

We have to design and implement a 4-bit universal shift register.

Excitation Table:

Input					Output				
Clear	Clock	S_1	S_0	Internal signal D_i ($i=0,1,2,3$)	A_3	A_2	A_1	A_0	Mode
0	X	X	X	X	0	0	0	0	Asynchronous Clear
1	X	0	0	X	A_3	A_2	A_1	A_0	Data Hold
1	$\uparrow$	0	1	A_{i-1}	I_{right}	A_3	A_2	A_1	Shift Right
1	$\uparrow$	1	0	A_{i+1}	A_2	A_1	A_0	I_{left}	Shift Left
1	$\uparrow$	1	1	A_i	I_3	I_2	I_1	I_0	Parallel Load

circuit Diagram:



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